MIL-R-2778B(SHIPS)
9 August 1965
SUPERSEDING
MIL-R-2778A(SHIPS)
22 April 1957
(See paragraph 6 2)

MILITARY SPECIFICATION

RUBBER SHEET, SOLID, UNVULCANIZED, HIGH GRAPHITE, GASKET USE, SYMBOL 2352

1. SCOPE

1.1 This specification covers unvulcanized rubber sheet with high graphite content for gasket use. The material is identified by symbol 2352.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY
MIL-P-116 - Preservation, Methods of.

STANDARDS

FEDERAL
FED-STD-601 - Rubber· Sampling and Testing.

MILITARY
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Copies may be obtained from the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N Y. 10016.)

3. REQUIREMENTS

3.1 Material.-

3.1.1 Composition - The composition of the unvulcanized rubber sheet compound shall be as specified in table I

3.1.2 Other ingredients - The remainder of the compound shall consist of ingredients which shall be free from any substance likely to have a deleterious effect on the compound. No rubber substitutes or reclaimed rubber shall be used in the compound.

FSC 5330

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Table I - Composition

Rubber (by volume) minimum	Organic-acetone extract 1/ (by weight)	Total sulfur 1/ (by weight)	Graphite (by weight)
Percent	Percent	Percent	Percent
40	15 to 30	10 to 20	24 to 32

1/ Based on the weight of the rubber, exclusive of sulfur contained in other ingredients.

3 2 Form.- Unless otherwise specified in the contract or purchase order, the unvulcanized rubber sheet material shall be furnished in rolls, 36 inches wide, with trimmed edges and weighing not less than 100 pounds. The material in the roll shall be adequately protected so that the layers will not stick together (see 6.1).

3.3 Weight - The weight of the rubber sheet 1/32 inch thick shall be not less than 2 nor more than 3 pounds per square yard, other thicknesses shall be in proportion.

3 4 Thickness and tolerance - Unless otherwise specified (see 6 1), the thickness of the rubber sheet material shall be 1/32 inch with a plus or minus tolerance of 0.008 inch.

3 5 Hardness - The hardness shall be not more than 80 durometer points when tested as specified in 4.7 1

3 6 Workmanship.- Workmanship shall be first class in every respect. The rubber sheet material shall be examined for defects as specified in 4.3 and shall be of uniform thickness and free of foreign matter and voids.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for examinations and tests.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all examinations and tests as specified herein. The supplier may utilize any laboratory acceptable to the Government Complete records of examinations and results of tests shall be available to the Government. The Government reserves the right to perform or observe any of the examinations or tests set forth in the specification where such examinations or tests are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.-

4 2 1 Lot - For the purposes of sampling. examination and tests, a lot shall consist of all sheet material of the same thickness. produced in one plant under essentially the same conditions, and offered for delivery at one time

4.2.2 Sampling for examination - A random sample of rolls shall be taken from each lot in accordance with the sampling plan in table II for the examination specified in 4 3. A unit area is defined as an area of one square yard and each unit area shall be taken at random from the rolls of material.

4.2 3 Sampling for tests - Random samples shall be taken from each lot that passes the examination of 4 3, in sufficient quantity to conduct either the production check tests or the quality conformance tests specified in 4 4 or 4 5 as applicable

4 3 Visual and dimensional examination - Each sample yard of rubber sheet material taken in accordance with 4 2 2 shall be examined for workmanship, dimensions, tolerances, weight, visual defects and any other requirement of this specification not covered by tests Any sample containing one or more defects shall be cause for rejection, and if the number of defects exceeds the applicable acceptance number in table II, this shall be cause for rejection of the entire lot

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Table II - Sampling for examination.

Lot size Number of unit areas	Sample size Number of unit areas	Number of defective unit areas of material	
		Acceptance number	Rejection number
Up to 8	all	0	1
9 to 25	8	1	2
26 to 90	13	2	3
91 to 150	20	3	4
151 to 280	32	5	6
281 to 500	50	7	8
501 to 1200	80	10	11
1201 to 3200	125	14	15
3201 and over	200	21	22

4.4 Production check tests.- Production check tests shall be conducted on samples from the first lot of material and from every tenth lot thereafter. All of the tests specified in 4.7 shall be conducted.

4 5 Quality conformance tests.- Quality conformance tests shall be conducted on samples from all intermediate lots on which production check tests were not performed The test specified in 4 7 1 shall be conducted.

4.6 Action in case of nonconformance - If any of the samples in the production check tests or quality conformance tests is found not to be in conformance with the requirements of this specification, this shall be cause for rejection of the lot represented by the sample. Furthermore, additional samples shall be taken or provided from each subsequent lot and shall be subjected to the test or tests wherein the failure occurred. Each lot shall then be accepted only after satisfactory results are obtained on the test or tests by all the samples taken or provided to represent the lot. This additional testing shall be discontinued after four successive lots have passed the test or tests.

4.7 Test methods.-

4.7.1 Hardness.- The hardness shall be determined in accordance with method 3021 of FED-STD-601 using plied up specimens and a shore A durometer. At least five readings shall be taken and the results shall be averaged. The material shall be conditioned at 27° plus or minus 5°C. (80°±9°F) for not less than 3 hours before determining the hardness.

4 7.2 Graphite content test.- The graphite content of the rubber sheet material shall be determined as follows: One-half gm. of the sample, cut into small pieces, shall be heated with 25 ml of cresol in an Erlenmeyer flask on a hot plate 10 to 12 hours, at approximately 375°F, with frequent shakings, cool, add 75 ml of benzol, filter through a weighed alundum crucible and wash thoroughly with benzol, dry to constant weight The residue in the crucible consists of graphite and other filters A complete chemical analysis shall then be made for all inorganic fillers, such as barium sulfate, zinc sulfide and similar inorganic fillers, and their combined weights subtracted from the weight of the residue in the crucible, and the difference taken as the weight of the graphite originally present in the sample.

4.7.3 Acetone extract test - The method of test for determining the acetone extract of the rubber sheet material shall be as specified in method 16321 of FED-STD-601.

4 7 4 Rubber and sulfur content tests - The rubber and sulfur content of the unvulcanized rubber sheet material shall be determined by methods 16121 and 16211 of FED-STD-601, respectively.

4.8 Examination of preparation for delivery - The packaging, packing, and marking shall be examined for compliance with section 5 of this document.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use.-

5.1.1 Preservation and packaging.- Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early material use.

5.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier, at the lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early material use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations.

5.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, lot number, date (month and year) of manufacture, size, contract or order number, supplier's name, and destination.

5.2 Onboard repair parts -

5.2.1 Preservation and packaging.- The rubber sheet material shall be rolled and securely wrapped in 60 pound minimum basis weight kraft paper and packaged in accordance with method III of MIL-P-116.

5.2.2 Packing - Packing of rubber sheet shall be in accordance with 5.1.2.

5.2.3 Marking.- In addition to any special marking required by the contract or order, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129. The information shall include nomenclature, Federal stock number or manufacturer's part number, lot number, date of manufacture, size, contract or order number, supplier's name, and destination.

5.3 Domestic shipment and storage or overseas shipment.- The requirements of levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.3.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.3.1.1 Preservation and packaging.-

5.3.1.1.1 Level A - The rubber sheet material shall be rolled and securely wrapped in 60 pound minimum basis weight kraft paper and packaged in accordance with method III of MIL-P-116.

5.3.1.1.2 Level C.- Preservation and packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity for early material use. This level may conform to the supplier's commercial practice when such meets these requirements.

5.3.1.2 Packing.-

5.3.1.2.1 Level A - The rubber sheet material, packaged as specified, shall packed in snug fitting, overseas type, wire-bound wood, wood-cleated fiberboard, cleated-plywood, nailed wood, fiberboard, or paper overlaid wood-cleated veneer boxes, conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-576, respectively, at the option of the supplier. Shipping containers shall have caseliners conforming to MIL-L-10547 and seams and closures shall conform to the appendix thereto. Caseliners may be omitted for boxes conforming to PPP-B-636 provided all joints and seams are sealed with waterproof tape. The boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto The gross weight of wood boxes shall not exceed 200 pounds, and the fiberboard boxes shall not exceed 70 pounds.

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5.3.1.2 2 Level B - The rubber sheet material, packaged as specified (see 6.1), shall be packed in snug fitting, domestic type, wire-bound wood, wood-cleated fiberboard, cleated-plywood, nailed wood, fiberboard, or paper overlaid wood-cleated veneer boxes conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-576, respectively, at the option of the supplier. Closures shall conform to applicable specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds, and of fiberboard boxes shall not exceed 70 pounds

5 3 1.3 Marking.- Marking for shipment shall be in accordance with 5.2 3)

6 NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Quantity of rubber sheet material required (see 3.2).
(c) Thickness of rubber sheet material required (see 3 4).
(d) Level of packaging and packing (see 5.3)
(e) Special marking, if required (see 5.2.3).

6 2 CHANGES FROM PREVIOUS ISSUE.- THE EXTENT OF CHANGES (DELETIONS. ADDITIONS, ETC) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT

Preparing activity
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(Project 5330-N069Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE D14000

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NOTICE OF VALIDATION

MIL-R-2778B(SHIPS)
NOTICE 2
27 March 1991

MILITARY SPECIFICATION

RUBBER SHEET, SOLID, UNVULCANIZED, HIGH GRAPHITE, GASKET USE, SYMBOL 2352

MIL-R-2778B(SHIPS), dated 9 August 1965, has been reviewed and determined to be valid for use in acquisition

Preparing activity.
Navy - SH

AMSC N/A FSC 5330

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